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Rakibul Hasan

Data Scientist | AI/ML Researcher

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Research Profile

Data Scientist and ML/AI Researcher with 3+ years of freelance data analysis experience and an MSc in Data Science from BUET. Research interests center on time-series forecasting, deep learning, Transformer architectures, and hybrid predictive models, with additional experience in machine learning, statistical analysis, and Big Data Analytics. Currently conducting MSc thesis research on deep learning and Transformer-based time-series forecasting, while developing practical expertise in PySpark, streaming analytics, data engineering, and cloud-based data platforms. Experienced in translating real-world datasets into reproducible analytical and predictive solutions, with research presented at international conferences.

Research Interests

Time-Series Forecasting | Deep Learning | Transformer Architectures | Hybrid Predictive Models | Machine Learning | Spatio-Temporal Modeling | Big Data Analytics | AI for Environmental Applications

Technical Skills

Programming & Data	Python (Pandas, NumPy, SciPy) | SQL | Jupyter Notebook
Statistics & Analytics	Exploratory Data Analysis | Statistical Analysis | Hypothesis Testing | Feature Engineering | Data Cleaning & Validation
Machine Learning	Regression | Classification | Clustering | Random Forest | XGBoost | LightGBM | SVM | Model Evaluation & Cross-Validation
Deep Learning & AI	TensorFlow | CNN | RNN | LSTM | GRU | Attention Mechanisms | Sequence-to-Sequence Models | Transformers
Time Series	ARIMA | SARIMA | SARIMAX | Multivariate Forecasting | Time-Series Decomposition | Trend & Seasonality Analysis | Spatio-Temporal Analysis
Big Data & Streaming	PySpark | Spark SQL | RDDs | MapReduce | Structured Streaming | Kafka | ETL Pipelines | Lakehouse Architecture
Data Engineering	Apache Airflow | Data Warehousing | Data Pipelines | Docker | GFS
Cloud & Platforms	Google Cloud Platform | Google Cloud Storage | BigQuery | Dataproc | Compute Engine
Visualization	Power BI | Matplotlib | Seaborn | Plotly | GeoPandas
Research	Research Methodology | Literature Review | Experimental Design | Academic Writing | LaTeX | Technical Documentation

Research Experience

Deep Learning & Transformer-Based Time-Series Forecasting **2025 – Present**
MSc Data Science Thesis – Bangladesh University of Engineering and Technology (BUET) *Dhaka, Bangladesh*

- Conduct research on multivariate time-series forecasting using environmental datasets, with emphasis on identifying temporal dependencies, trends, seasonality, and non-stationarity.
- Investigate deep learning and Transformer-based architectures for long-horizon forecasting and representation of complex temporal dependencies.
- Developing a hybrid forecasting framework that combines complementary modeling approaches to improve predictive performance and robustness.
- Evaluate forecasting models using systematic experimental design and comparative performance analysis across multiple datasets and forecasting horizons.

Intelligent Spark Configuration Optimization **2026**
Big Data Analytics / Research Project

- Investigate data-driven Spark configuration optimization using workload characteristics, performance metrics, and machine-learning-based optimization strategies.
- Explore Gaussian Process modeling and Bayesian optimization for selecting promising configurations within a constrained search space.

Selected Projects

Route-Level Pricing Bias Analysis **2025**
Independent Research – ICASDS 2025, Institute of Statistical Research and Training, University of Dhaka

- Analyzed large-scale taxi trip data to identify route-level pricing disparities using statistical analysis and machine learning.
- Developed a hybrid fairness metric to quantify pricing bias and interpret differences in fare patterns across routes.

Hybrid Model For World Monthly CO2 Forecasting **2026**
Data Mining Project

- Applied time-series analysis and machine learning to model and forecast carbon dioxide emissions for data-driven environmental awareness and mitigation planning.
- Compared forecasting approaches and analyzed temporal patterns to generate interpretable predictive insights.

Camera Fingerprinting for AI-Generated Image Detection **2025**
Deep Learning Research Project

- Investigated camera sensor noise fingerprints as a physical signal for distinguishing authentic photographs from AI-generated images.
- Achieved 0.715 test accuracy and 0.80 AUC in experimental evaluation.

Image Caption Generation with CNN-LSTM **2024**
Deep Learning Project

- Developed an encoder-decoder image captioning model combining CNN-based visual feature extraction with LSTM-based sequence generation.
- Achieved BLEU-1 and BLEU-2 scores of 0.403 and 0.205, respectively.

Selected Publications

- **Forecasting Carbon Dioxide Emissions: A Data Science Approach to Awareness and Mitigation** – ICCESD 2026, Khulna University of Engineering & Technology (KUET), Bangladesh.
- **Modeling and Forecasting Border Mobility at U.S. Entry Points: A Data-Driven Transportation Engineering Approach** – ICCERI 2025, Rajshahi University of Engineering & Technology (RUET), Bangladesh.
- **Behind the Meter: A Machine Learning Framework for Taxi Fare Fairness with a Hybrid Metric to Identify Route-Level Pricing Bias** – ICASDS 2025, Institute of Statistical Research and Training, University of Dhaka.

Professional Experience

Freelance Data Analyst **Oct 2022 – Oct 2024**
Fiverr (Remote) *Dhaka, Bangladesh*

- Delivered data analysis and visualization solutions for international clients across diverse real-world datasets and business domains.
- Cleaned, transformed, validated, and analyzed datasets using Python, SQL, and Excel to identify trends, anomalies, and actionable insights.
- Developed analytical reports and interactive visualizations to communicate findings clearly to technical and non-technical stakeholders.
- Collaborated with clients to translate analytical requirements into reproducible data-driven solutions and decision-ready deliverables.

Education

Master of Science, Data Science *Bangladesh University of Engineering and Technology (BUET)* Sept 2024 – Ongoing
Coursework completed; thesis ongoing.
Relevant Coursework: Big Data Analytics, Data Mining, Machine Learning, Advanced ML, Deep Learning, AI, NLP

Bachelor of Science, Physics *National University – Govt. Haraganga College* Aug 2017 – Dec 2023
CGPA: 3.37/4.00

Certifications

Google Data Analytics Professional Certificate (Coursera) | Associate Data Scientist (DataCamp) | SQL for Data Science (Coursera) | Data Analysis with R Programming (Coursera)